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Predictive models for diabetes mellitus using machine learning techniques

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Abstract

Background: Diabetes Mellitus is an increasingly prevalent chronic disease characterized by the body's inability to metabolize glucose. The objective of this study was to build an effective predictive model with high sensitivity and selectivity to better identify Canadian patients at risk of having Diabetes Mellitus based on patient demographic data and the laboratory results during their visits to medical facilities.

Methods: Using the most recent records of 13,309 Canadian patients aged between 18 and 90 years, along with their laboratory information (age, sex, fasting blood glucose, body mass index, high-density lipoprotein, triglycerides, blood pressure, and low-density lipoprotein), we built predictive models using Logistic Regression and Gradient Boosting Machine (GBM) techniques. The area under the receiver operating characteristic curve (AROC) was used to evaluate the discriminatory capability of these models. We used the adjusted threshold method and the class weight method to improve sensitivity - the proportion of Diabetes Mellitus patients correctly predicted by the model. We also compared these models to other learning machine techniques such as Decision Tree and Random Forest.

Results: The AROC for the proposed GBM model is 84.7% with a sensitivity of 71.6% and the AROC for the proposed Logistic Regression model is 84.0% with a sensitivity of 73.4%. The GBM and Logistic Regression models perform better than the Random Forest and Decision Tree models.

Conclusions: The ability of our model to predict patients with Diabetes using some commonly used lab results is high with satisfactory sensitivity. These models can be built into an online computer program to help physicians in predicting patients with future occurrence of diabetes and providing necessary preventive interventions. The model is developed and validated on the Canadian population which is more specific and powerful to apply on Canadian patients than existing models developed from US or other populations. Fasting blood glucose, body mass index, high-density lipoprotein, and triglycerides were the most important predictors in these models.

Keywords: Diabetes mellitus; Gradient boosting machine; Machine learning; Misclassification cost; Predictive models.

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Figures



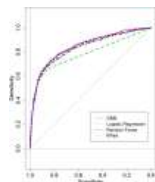


Fig. 1 Information gain measure from predictors

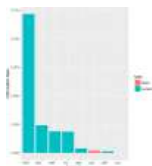


Fig. 2 Receiver operating curves for the...

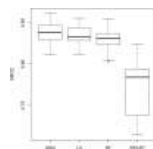


Fig. 3 Box plot: comparing the AROC...

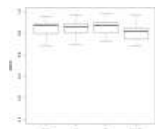


Fig. 4 Box plot of AROC values...



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